Membership Inference Attacks From First Principles

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Background: Membership Inference Attacks

- **Goal:** Given a model f_θ and a sample (x, y), determine whether it was in the training set.
- This helps us check whether a model reveals information about its training data.
- Existing works evaluate attacks using **average-case metrics**
  - accuracy
  - ROC-AUC

Problem

These metrics fail to capture whether an attack can confidently identify even a few training members.

Why Focus on Low FPR?

- In membership inference, predicting "member" is a strong claim
  - a false positive means wrongly claiming a user was in the training data
- A useful attack should make **very few such mistakes**
- Therefore, we evaluate:
  - how many true members can be identified
  - when false positives are extremely rare
- This corresponds to measuring:

TPR at very low FPR (e.g., 0.1%)

Key Contributions of the Paper

1. **A new evaluation principle**
   - Focus on **TPR at low FPR**
2. **Reformulation as hypothesis testing**
   - Decide whether (x, y) was used in training or not
3. **Per-example distribution modeling**
   - Estimate:

$$Q_{\text{in}}(x, y), \quad Q_{\text{out}}(x, y)$$

4. **Likelihood Ratio Attack (LiRA)**
   - Compare how likely the model output $f(x)$ is under Q_{in} and Q_{out}, and choose the more likely one

Result: *Up to* $10\times$ *stronger at low false-positive rates*

Membership Inference: Setup

- A model is trained on a dataset D
- Given a sample (x, y), we want to decide:

$$\text{Is } (x, y) \in D \text{ ?}$$

- We observe the model output $f(x)$
- Possibly query the model multiple times

Typical approach:

$$\text{score}(x, y) \gtrless \tau$$

A Simple Baseline: LOSS Attack

- Idea: training samples tend to have lower loss

$$\text{score}(x, y) = -\ell(f(x), y)$$

$$\text{Predict member if score}(x, y) > \tau$$

Problem:

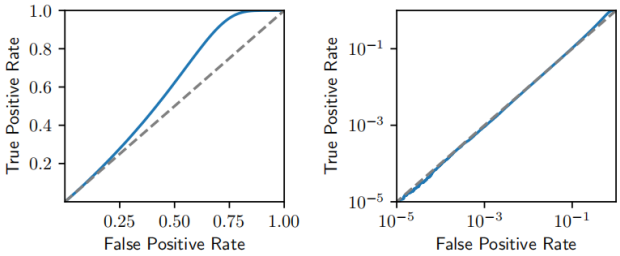
- Uses a **single global threshold**
- Assumes all samples behave similarly

Why the LOSS Baseline Fails

- LOSS assumes:

$$\ell(f(x), y) \text{ is comparable across all samples}$$

- In reality:
  - some samples are easier or harder
  - loss distributions vary across samples
- As a result:
  - performance may look good on a linear scale
  - but fails in the low-FPR regime



Membership Inference as Hypothesis Testing

- Goal: decide whether (x, y) was used in training

$$\begin{cases} H_1 : (x, y) \text{ is in training set} \\ H_0 : (x, y) \text{ is not in training set} \end{cases}$$

- Model output follows different distributions:

$$f(x) \sim \begin{cases} Q_{\text{in}} & \text{if } H_1 \\ Q_{\text{out}} & \text{if } H_0 \end{cases}$$

- Compare likelihood under the two distributions:

$$\Lambda(x, y) = \frac{p(f(x) \mid Q_{\text{in}})}{p(f(x) \mid Q_{\text{out}})}$$

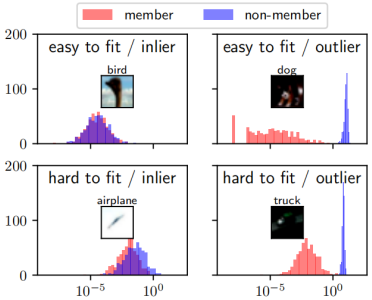
- If Λ is large $\Rightarrow$ predict "member"

Why Per-Example Modeling?

- Different samples behave very differently

$$p(f(x) \mid Q_{\text{in}}), \quad p(f(x) \mid Q_{\text{out}})$$

- depend on the specific sample (x, y)
- Therefore:
  - estimate distributions **per example**
  - instead of using a global threshold



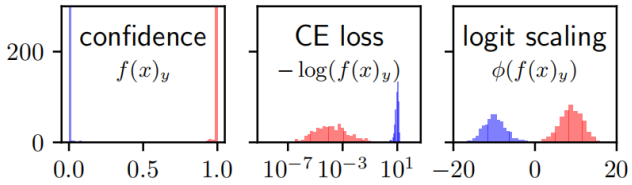
Gaussian Modeling and Logit Transform

- Approximate distributions with Gaussians:

$$Q_{\text{in}} \approx \mathcal{N}(\mu_{\text{in}}, \sigma^2_{\text{in}}), \quad Q_{\text{out}} \approx \mathcal{N}(\mu_{\text{out}}, \sigma^2_{\text{out}})$$

- Use model confidence $p = f(x)_y$

$$\phi(p) = \log\left(\frac{p}{1-p}\right)$$



Online LiRA

- For a target example (x, y), train *shadow models* on random datasets.
- Some shadow models are trained **with** (x, y) (IN), others **without** (x, y) (OUT).
- Compute the transformed confidence $\phi(f(x)_y)$ for both groups.
- Fit two Gaussians:

$$\mathcal{N}(\mu_{\text{in}}, \sigma^2_{\text{in}}), \qquad \mathcal{N}(\mu_{\text{out}}, \sigma^2_{\text{out}})$$

- Evaluate the observed target-model score through a likelihood ratio.

Algorithm 1 Our online Likelihood Ratio Attack (LiRA). We train shadow models on datasets with and without the target example, estimate mean and variance of the loss distributions, and compute a likelihood ratio test. (In our **offline** variant, we omit lines 5, 6, 10, and 12, and instead return the prediction by estimating a single-tailed distribution, as is shown in Equation (4).)

Require: model f, example (x, y), data distribution $\mathbb{D}$

1: $\text{confs}_{\text{in}} = \{\}$
2: $\text{confs}_{\text{out}} = \{\}$
3: **for** N times **do**
4: $\quad D_{\text{attack}} \leftarrow^{\$} \mathbb{D}$ ▷ *Sample a shadow dataset*
5: $\quad f_{\text{in}} \leftarrow \mathcal{T}(D_{\text{attack}} \cup \{(x, y)\})$ ▷ *train IN model*
6: $\quad \text{confs}_{\text{in}} \leftarrow \text{confs}_{\text{in}} \cup \{\phi(f_{\text{in}}(x)_y)\}$
7: $\quad f_{\text{out}} \leftarrow \mathcal{T}(D_{\text{attack}} \backslash \{(x, y)\})$ ▷ *train OUT model*
8: $\quad \text{confs}_{\text{out}} \leftarrow \text{confs}_{\text{out}} \cup \{\phi(f_{\text{out}}(x)_y)\}$
9: **end for**
10: $\mu_{\text{in}} \leftarrow \texttt{mean}(\text{confs}_{\text{in}})$
11: $\mu_{\text{out}} \leftarrow \texttt{mean}(\text{confs}_{\text{out}})$
12: $\sigma^2_{\text{in}} \leftarrow \texttt{var}(\text{confs}_{\text{in}})$
13: $\sigma^2_{\text{out}} \leftarrow \texttt{var}(\text{confs}_{\text{out}})$
14: $\text{conf}_{\text{obs}} = \phi(f(x)_y)$ ▷ *query target model*
15: **return** $\Lambda = \dfrac{p(\text{conf}_{\text{obs}} \mid \mathcal{N}(\mu_{\text{in}}, \sigma^2_{\text{in}}))}{p(\text{conf}_{\text{obs}} \mid \mathcal{N}(\mu_{\text{out}}, \sigma^2_{\text{out}}))}$

Offline LiRA

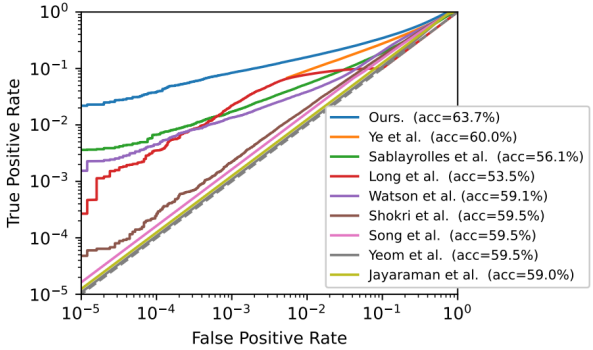
- Online LiRA is strong, but expensive: it requires training new shadow models after the query example is known.
- The offline variant trains only **OUT** shadow models in advance.
- It then performs a one-sided test relative to the OUT distribution:

$$\Lambda = 1 - \Pr[Z > \phi(f(x)_y)], \quad Z \sim \mathcal{N}(\mu_{\text{out}}, \sigma^2_{\text{out}}).$$

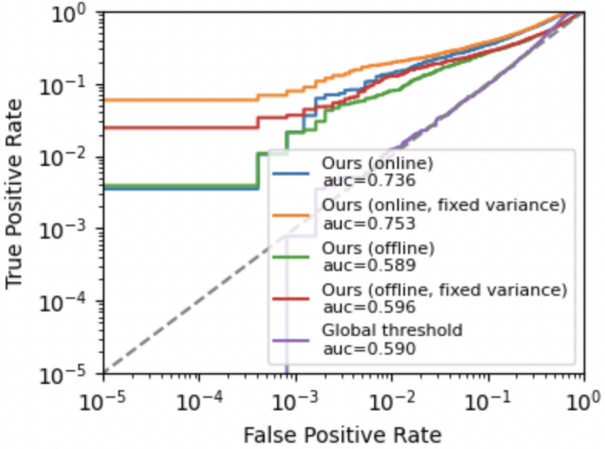
- This is much easier to deploy, while keeping performance relatively close to the online version.

Main Result: Performance at Low FPR

- Standard metrics (accuracy, AUC) can be misleading
- At very low FPR:
  - many prior attacks almost completely fail
  - LiRA remains effective
- The paper shows improvements of up to $\mathbf{10}\times$



Our reproduced results (CIFAR-10, WideResNet-28-2)



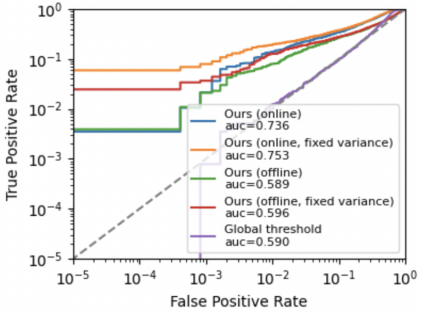
Summary at FPR = 0.1%

Variant	AUC	Acc.	TPR
Online	73.56	63.85	2.23
Online (fixed var.)	75.32	67.34	7.91
Offline	58.93	58.37	2.21
Offline (fixed var.)	59.58	58.87	3.56
Global threshold	58.98	59.72	0.09

- Online variants are clearly stronger than the global-threshold baseline.
- In our experiment, the fixed-variance online variant performs best.

Interpretation of Our Reproduction

- Global-threshold methods almost fail at low FPR
  - TPR is only **0.09**
- Offline LiRA remain effective in this regime
- Evaluating at low FPR fundamentally changes the ranking of attacks
- Modeling score distributions is more effective than using global thresholds



Limitations

- Requires data from a similar distribution to train shadow models
- Online LiRA can be computationally expensive
- Gaussian assumption is approximate
- Performance depends on model architecture and query access

Conclusion

1. Privacy attacks should be evaluated in the **low false-positive regime**.
2. LiRA reframes membership inference as a likelihood-ratio test with:
   - per-example hardness,
   - logit-scaled confidence,
   - and Gaussian modeling.
3. Both the paper and our reproduction show that this leads to much stronger attacks than older baselines in the regime that matters most.

Thank you!